$n \times n$ Cyclic Latin Squares representing solutions of the (n, c)-Coins Problem for $0 \leq c \leq n$

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Abstract

$\forall n \in \mathbb{N},\ n > 0,\ n \notin \{2, 3, 4, 6\}$, $\exists p : \{0, \ldots n-1\} \to \{0, \ldots n-1\}$, a permutation of $\{0, \ldots n-1\}$, such that $\forall i, j \in \{0, \ldots n-1\}$, the square with cells $S_{i,j} = (i + p(j)) \bmod n$ is an $n \times n$ *Cyclic Latin Square*, representing solutions of (n, c)-*Coins Problems* $\forall c\ \ 0 \leq c \leq n$. An (n, c)-Coins Problem for $c = 1$ corresponds to an n-*Queens Problem*.

We have an *existence proof* in `Lean` and corresponding *construction code* in `Haskell`, `Julia` and `Lean`.

See GitHub and also MO#510135, PSE#138939, MO#516225.

The proof uses the *Injective-Majorization Principle* that, instead of intensive checking inequalities on each diagonal, uses the existence of a single function $f(i, j) \leq S_{i,j}$ that is injective on each diagonal.

The proof uses four subsets of $\mathbb{N}$ covering almost all $\mathbb{N}\, mod\, 6$ classes :

- $n \equiv 1, 5 \pmod 6$ (with $n \geq 1$),
- $n \equiv 2, 4 \pmod 6$ (with $n \geq 10$),
- $n \equiv 3 \pmod 6$ (with $n \geq 27$),
- $n \equiv 0 \pmod 6$ (with $n \geq 12$).

For all four subsets a Cyclic Latin Square can be constructed collectively for all their elements.

Constructing a Cyclic Latin Square for the four uncovered natural numbers, $n \in \{8, 9, 15, 21\}$, is handled individually.

Likewise, constructing a (non-Cyclic) Latin Square for the two remaining natural numbers, $n \in \{4, 6\}$, is also handled individually.

1 Cyclic Latin Squares

1.1 From (n, c)-Coins Problems to Latin Squares

Given $n > 0$, and $c \in \{0, \ldots, n\}$ solutions of (n, c)-Coins Problems are subsets S_c of $n.c$ cells of an $n \times n$ square $S_{i,j}$ such that every row, column, s-diagonal $(i + j = s)$, and d-diagonal $(i - j = d)$, contains at most c cells of S_c.

The construction of Le, Li and Wang [1] constructs *one* S_c for each c.

Our construction constructs an $n \times n$ Cyclic Latin Square $S_{i,j}$ representing *many* S_c's at once using $S_c = \{(i, j) \mid S_{i,j} \leq c\}\ \ \forall c\ \ 0 \leq c \leq n$.

2 Theorem Statement

Theorem 2.1. *$\forall n \in \mathbb{N},\ n > 0,\ n \notin \{2, 3, 4, 6\}$, $\exists p : \{0, \dots n-1\} \to \{0, \dots n-1\}$, a permutation, such that $S_{i,j} = (i + p(j)) \bmod n$ is a Cyclic Latin Square satisfying :*

(i) $\forall i \in \{0, \dots n-1\}$, the row map $j \mapsto S_{i,j}$ is bijective.
(ii) $\forall j \in \{0, \dots n-1\}$, the column map $i \mapsto S_{i,j}$ is bijective.
(iii) $\forall k : 0 \leq k < n, \forall s \in \mathbb{N}$, $|\{(i,j) \in s\text{-diagonal with } S_{i,j} \leq k\}| \leq k+1$.
(iv) $\forall k : 0 \leq k < n, \forall d \in \mathbb{Z}$, $|\{(i,j) \in d\text{-diagonal with } S_{i,j} \leq k\}| \leq k+1$.

3 Injective Majorization

Instead of using direct cardinality counting for (iii) and (iv) above, which would computationally explode for large n, we use a principle that reduces the problem for *all* thresholds $0 \leq k < n$ simultaneously to constructing a single dominated injection.

In what follows, unless explicitely stated, diagonal refers to either s-diagonal or d-diagonal.

3.1 The Injective-Majorization Principle

Let $S_{i,j}$ with $0 \leq i, j < n$ be a square and let D be any diagonal.

Lemma 3.1. *If there exists a function $f : \mathbb{N} \times \mathbb{N} \to \mathbb{N}$ such that:*

(1) Majorization : $\forall i, j : 0 \leq i, j < n, f(i,j) \leq S_{i,j}$.
(2) Diagonal Injectivity : f is injective on D.

Then for every threshold $k \in \mathbb{N}$, the set $D_{\leq k} = \{(i,j) \in D \mid S_{i,j} \leq k\}$ has cardinality at most $k+1$.

Proof. By condition (1), $0 \leq f(i,j) \leq S_{i,j} \leq k$ so $f(i,j) \in \{0, \dots, k\}$. By condition (2), the restriction of f to $D_{\leq k}$ is an injective map into $\{0, \dots, k\}$. Therefore $|D_{\leq k}| \leq |\{0, \dots, k\}| = k+1$.

□

4 Theorem Proof

Recall that $i = s - j$ on every s-diagonal and $i = d + j$ on every d-diagonal.

The explanations below will use those equalities without explicitly mentioning them.

Recall that the four families are

- Family 1 : $n \equiv 1, 5 \pmod 6$ with $n \geq 1$,
- Family 2 : $n \equiv 2, 4 \pmod 6$ with $n \geq 10$,
- Family 3 : $n \equiv 3 \pmod 6$ with $n \geq 27$,
- Family 4 : $n \equiv 0 \pmod 6$ with $n \geq 12$.

and notice that

- Family 2 does not include $n = 8$.
- Family 3 does not include $n = 9$, $n = 15$ and $n = 21$.

Those cases are handled separately.

We will define p and f and prove that

- p is a bijection on $\{0, \dots, n-1\}$,
- f is an injection on the diagonals involved,
- $f(i,j) \leq S_i j$ on the diagonals involved.

4.1 Family 1

Let $p(j) = (-2j) \bmod n$.

p is bijective since $\gcd(n, 2) = 1$.

On an s-diagonal, $S_{i,j} = (s - 3j) \bmod n$ is injective since $\gcd(3, n) = 1$ (see Example 4.1 and Figure 1).

On a d-diagonal, $S_{i,j} = (d - j) \bmod n$ is trivially injective (see Example 4.2 and Figure 2).

Let $f(i, j) = S_{i,j}$.

4.2 Family 2

Let $m = n/2$.

Let $p(j) = (2j + m + 1) \bmod n$ for $0 \le j < m$.
Let $p(j) = (2(j - m) + m + 4) \bmod n$ for $m \le j < n$.

p is bijective since the former covers all $(m + 1) \bmod 2$ values one-to-one , and the latter covers all $m \bmod 2$ values one-to-one.

On an s-diagonal, inside each half, $S_{i,j} = (s + j + m + 1) \bmod n$ has step $+1$. Because the second half is shifted by $3 \bmod n$, a collision between $j_1 < m \le j_2$ happens when $j_2 - j_1 = n - 3$. This happens only when $j_1 \in \{0, 1, 2\}$ (and $i_1 \in \{n-3, n-2, n-1\}$), the 3×3 bottom-left corner, and $j_2 \in \{n-3, n-2, n-1\}$ (and $i_2 \in \{0, 1, 2\}$), the 3×3 top-right corner.

Let $f(i, j) = j - (n - 3) \in \{0, 1, 2\}$ on the 3×3 top-right corner. One of the values, skipped by the $+3$ shift, are vacant on each s-diagonal passing through it (see Example 4.3 and Figure 3). Trivially $f(i, j) \le S_{i,j}$.

In the 3×3 top-right corner ($j = n - 3 + \delta$ with $\delta \in \{0, 1, 2\}$), $p(j) = m - 2 + 2\delta$, so the largest sum $i + p(j)$ occurs at $(i, j) = (2, n - 1)$, where $2 + p(n - 1) = m + 4$. The threshold $n \ge 10$ ($m \ge 5$) ensures $m + 4 < 2m = n$, so $S_{i,j}$ never wraps around $\bmod n$.

On a d-diagonal, inside each half, $S_{i,j} = s + 3j - m + 4$ has step $+3$. Because the second half is shifted by $3 \bmod n$, a collision between $j_1 < m \le j_2$ happens when $j_2 - j_1 = n - 1$, equivalent with $(j_2 - j_1) + 1 \equiv 0 \pmod n$, equivalent with $3(j_2 - j_1) + 3 \equiv 0 \pmod n$ since $\gcd(3, n) = 1$. This happens only at a single pair of cells on the entire board: $(0, 0)$ and $(n - 1, n - 1)$ on the main diagonal.

Let $f(n - 1, n - 1) = 1$, one of the values, skipped by the $+3$ shift, which is vacant on the main diagonal (see Example 4.4 and Figure 4).

4.3 Family 3

Let $k = n/3$.

Let $p(j) = (3j + 2k + 4u) \bmod n$ with $u = \lfloor j/k \rfloor \in \{0, 1, 2\}$. $\{0, \ldots, n - 1\}$ is partitioned into three blocks of length k with $+4$ jumps between consecutive blocks.

p is bijective since block u covers residue class $(2k + u) \bmod 3$, and thus all three residue classes $\bmod 3$ are, one-to-one, covered.

On an s-diagonal, $S_{i,j} = (s + 2j + 2k + 4u) \bmod n$ has step $+2$ inside each block. At each of the two block transitions ($u = 0 \to 1$ and $u = 1 \to 2$), the extra $+4$ jump skips two steps of $+2$: $(s + k) \bmod n$ and $(s + k + 2) \bmod n$ at $j = k$, and $(s + 4) \bmod n$ and $(s + 6) \bmod n$ at $j = 2k$. Collisions between blocks $u_1 < u_2$ happen when $2(j_2 - j_1) + 4(u_2 - u_1) \equiv 0 \pmod n$, which is equivalent with $(j_2 - j_1) + 2(u_2 - u_1) = n$ since $\gcd(2, n) = 1$, forcing $u_1 = 0$, $u_2 = 2$ and $j_2 - j_1 = n - 4$. This happens only in the 4×4 top-right corner.

Let $f(i, j) = S_{i,j} - k \in \{s + k, s + k + 2\} \bmod n$ for $j \in \{n - 4, n - 3\}$, which fills the two vacant slots skipped at the $u = 0 \to 1$ transition ($j = k$) (see Example 4.5 and Figure 5).

Let $f(i,j) = S_{i,j} - 2k \in \{s+4, s+6\} \bmod n$ for $j \in \{n-2, n-1\}$, which fills the two vacant slots skipped at the $u = 1 \to 2$ transition ($j = 2k$) (see Example 4.5 and Figure 5).

In the 4×4 top-right corner, the largest unreduced sum $i + p(j)$ occurs at $(i,j) = (3, n-1)$, where $p(n-1) = 2k+5$ and $3 + p(n-1) = 2k+8$. The threshold $n \geq 27$ ($k \geq 9$) ensures $2k+8 < 3k = n$, so $S_{3,n-1} = 2k+8$ does not wrap around $\bmod n$ to a value $< 2k$ before subtracting $2k$ in $f(3, n-1) = S_{3,n-1} - 2k$.

On a d-diagonal, $S_{i,j} = (d + 4j + 2k + 4u) \bmod n$ has step $+4$ inside each block. At each of the two block transitions ($u = 0 \to 1$ and $u = 1 \to 2$), the extra $+4$ jump skips one step of $+4$: $d \bmod n$ at $j = k$, and $(d+k+4) \bmod n$ at $j = 2k$. Collisions between blocks $u_1 < u_2$ happen when $4(j_2 - j_1) + 4(u_2 - u_1) \equiv 0 \pmod n$, which is equivalent with $(j_2 - j_1) + (u_2 - u_1) = n$ since $\gcd(4, n) = 1$, forcing $u_1 = 0$, $u_2 = 2$ and $j_2 - j_1 = n - 2$. This happens only in the 2×2 bottom-right corner.

Let $f(i,j) = S_{i,j} - 2k = i + 2 \equiv d \pmod n$ for $j = n-2$, which fills the vacant slot skipped at the $u = 0 \to 1$ transition ($j = k$) (see Example 4.6 and Figure 6).
Let $f(i,j) = S_{i,j} - k = i + k + 5 \equiv d + k + 4 \pmod n$ for $j = n-1$, which fills the vacant slot skipped at the $u = 1 \to 2$ transition ($j = 2k$) (see Example 4.6 and Figure 6).

4.4 Family 4

Let $m = n/2$ and $k = n/3$.

Let $p(0) = n-2$, $p(m-2) = 0$ and $p(n-1) = 2$.
Let $p(j) = 2j + 2$ for $1 \leq j \leq m-3$.
Let $p(j) = 2(j - (m-1)) + 1$ for $m-1 \leq j \leq n-2$.

p is bijective since the first ones cover $\{0, 2, n-2\}$, and the next ones cover $\{4, 6, \ldots, n-4\}$, and the last ones cover all odd numbers $\{1, 3, \ldots, n-1\}$

The threshold $n \geq 12$ ($m \geq 6$ and $k \geq 4$) covers all valid multiples of 6 since $n = 6$ is excluded, and ensures that the first main block $1 \leq j \leq m-3$ is non-empty and that the column $j = n-5$ lies inside the second main block $m-1 \leq j \leq n-2$.

On an s-diagonal, inside the first main block $S_{i,j} = (s + j + 2) \bmod n$ has step $+1$, and inside the second main block $S_{i,j} = (s + j + 3) \bmod n$ also has step $+1$, leaving the two values $(s+m) \bmod n$ and $(s+m+1) \bmod n$ vacant between the two blocks. Collisions only occur with the three boundary pins $j \in \{0, m-2, n-1\}$.

At $j = m-1$ (for $i < n-1$): $S_{i,m-1} = i+1$ collides with pin $j = m-2$ ($S_{i+1,m-2} = i+1$), so let $f(i, m-1) = S_{i,m-1} - 1 = i \equiv s + m + 1 \pmod n$ (see Example 4.7 and Figure 7).
At $j = n-5$ (for $i + j < n$, so $i < 5$): $S_{i,n-5} = s-2$ collides with pin $j = 0$ ($S_{s,0} = s-2$), so let $f(i, n-5) = S_{i,n-5} - (m-2) = i + m - 5 \equiv s + m \pmod n$ (see Example 4.7 and Figure 7).
At $j = n-1$ (for $i < 2$): $S_{i,n-1} = i+2$ collides with $j = 1$ ($S_{s-1,1} = i+2$), while pin $j = 0$ is off the diagonal ($s \geq n-1$), leaving $i+1$ vacant; so let $f(i, n-1) = S_{i,n-1} - 1 = i + 1$ (see Example 4.7 and Figure 7).

On a d-diagonal, inside both main blocks $i + p(j)$, $S_{i,j} = (d + 3j + 2) \bmod n$ has step $+3$: $S_{i,j} = (d + 3j + 2) \bmod n$ in the first main block and $S_{i,j} = (d + 3j + 3) \bmod n$ in the second main block. Since $3 \mid n$ ($\gcd(3, n) = 3$), the first main block stays entirely in residue class $(d+2) \bmod 3$ and the second main block stays entirely in residue class $d \bmod 3$, leaving the third residue class $(d+1) \bmod 3$ completely vacant (except for the single pin $j = 0$). Within each block, a wrap-around collision $3(j_2 - j_1) = n$ occurs when $j_2 - j_1 = n/3 = k$ (for $i_2 \geq k$). To resolve these collisions (see Example 4.8 and Figure 8):

- In the first main block ($1 + k \leq j \leq m-3$ and $i \geq k$), let $f(i,j) = S_{i,j} - 1 \equiv d + 1 \pmod 3$.
- In the second main block ($m + k \leq j \leq n-2$ and $i \geq k$), let $f(i,j) = S_{i,j} - 2 \equiv d + 1 \pmod 3$.
- At the boundary column $j = m - 1 + k$, let $f(i,j) = S_{i,j} - 1$ for $k \leq i < 2k$, and $f(i,j) = S_{i,j} - 5$ (or $S_{i,j} - 4$ when $n = 12$) for $i \geq 2k$ to avoid colliding with the pins $j = m-2$ and $j = 0$.

4.5 Concrete Examples of Diagonal Slot Stealing

We present eight concrete examples—one s-diagonal and one d-diagonal for each of the four families—illustrated in Figures 1–8. In each grid, off-diagonal entries and grid lines are shown in gray, uncollided diagonal entries are highlighted in blue, the cell that keeps its original value in a collision pair is highlighted in orange (**v**), and the duplicate cell that steals a lower vacant slot is highlighted in red (**v** ↓ **f**).

Example 4.1 (Family 1 s-Diagonal: $n = 7$, $s = 5$). For $n = 7 \equiv 1 \pmod 6$, $p(j) = (-2j) \bmod 7$ gives $p = (0, 5, 3, 1, 6, 4, 2)$. Along the s-diagonal $i + j = 5$, $S_{i,j} = (5 - 3j) \bmod 7$ steps by $-3 \equiv +4 \pmod 7$, visiting $\{5, 2, 6, 3, 0, 4\}$, which are already distinct because $\gcd(3, 7) = 1$. Thus $f(i, j) = S_{i,j}$ requires no shifts (Figure 1).

	0	1	2	3	4	5	6
0	0	5	3	1	6	**4**	2
1	1	6	4	2	**0**	5	3
2	2	0	5	**3**	1	6	4
3	3	1	**6**	4	2	0	5
4	4	**2**	0	5	3	1	6
5	**5**	3	1	6	4	2	0
6	6	4	2	0	5	3	1

Cell (i, j)	$S_{i,j}$	$f(i, j)$	Status
$(5, 0)$	5	5	Distinct
$(4, 1)$	2	2	Distinct
$(3, 2)$	6	6	Distinct
$(2, 3)$	3	3	Distinct
$(1, 4)$	0	0	Distinct
$(0, 5)$	4	4	Distinct

Figure 1: Family 1 ($n = 7$): s-diagonal $s = 5$ (step -3 mod 7), already collision-free with $f(i, j) = S_{i,j}$.

Example 4.2 (Family 1 d-Diagonal: $n = 7$, $d = -1$). Along the d-diagonal $i - j = -1$ ($i = j - 1$) for $n = 7$, $S_{i,j} = (-1 - j) \bmod 7$ steps by $-1 \pmod 7$, visiting $\{5, 4, 3, 2, 1, 0\}$, which are trivially distinct. Thus $f(i, j) = S_{i,j}$ requires no shifts (Figure 2).

	0	1	2	3	4	5	6
0	0	**5**	3	1	6	4	2
1	1	6	**4**	2	0	5	3
2	2	0	5	**3**	1	6	4
3	3	1	6	4	**2**	0	5
4	4	2	0	5	3	**1**	6
5	5	3	1	6	4	2	**0**
6	6	4	2	0	5	3	1

Cell (i, j)	$S_{i,j}$	$f(i, j)$	Status
$(0, 1)$	5	5	Distinct
$(1, 2)$	4	4	Distinct
$(2, 3)$	3	3	Distinct
$(3, 4)$	2	2	Distinct
$(4, 5)$	1	1	Distinct
$(5, 6)$	0	0	Distinct

Figure 2: Family 1 ($n = 7$): d-diagonal $d = -1$ (step -1 mod 7), already collision-free with $f(i, j) = S_{i,j}$.

Example 4.3 (Family 2 s-Diagonal Piecewise Jump: $n = 10$, $s = 7$). For $n = 10$ ($m = 5$), $p = (6, 8, 0, 2, 4 \mid 9, 1, 3, 5, 7)$ has two linear halves ($j < 5$ and $j \geq 5$). Along $i + j = 7$, cells $(7, 0)$ in the first half and $(0, 7)$ in the 3×3 top-right corner ($j_2 - j_1 = 7 = n - 3$) both evaluate to $S_{7,0} = S_{0,7} = 3$. The values on the diagonal are $\{1, 2, \mathbf{3}, 4, 5, 6, 7\}$, leaving $\mathbf{0}$ vacant. Setting $f(0, 7) = j_2 - (n - 3) = 7 - 7 = 0 < 3$ resolves the collision (Figure 3).

	0	1	2	3	4	5	6	7	8	9
0	6	8	0	2	4	9	1	**3** ↓0	5	7
1	7	9	1	3	5	0	**2**	4	6	8
2	8	0	2	4	6	**1**	3	5	7	9
3	9	1	3	5	**7**	2	4	6	8	0
4	0	2	4	**6**	8	3	5	7	9	1
5	1	3	**5**	7	9	4	6	8	0	2
6	2	**4**	6	8	0	5	7	9	1	3
7	**3**	5	7	9	1	6	8	0	2	4
8	4	6	8	0	2	7	9	1	3	5
9	5	7	9	1	3	8	0	2	4	6

Cell (i, j)	$S_{i,j}$	$f(i, j)$	Status
$(7, 0)$	3	3	Keeps value
$(6, 1)$	4	4	Distinct
$(5, 2)$	5	5	Distinct
$(4, 3)$	6	6	Distinct
$(3, 4)$	7	7	Distinct
$(2, 5)$	1	1	Distinct
$(1, 6)$	2	2	Distinct
$(0, 7)$	3	**0**	**Steals vacant 0 (< 3)**

Figure 3: Family 2 ($n = 10$): s-diagonal $s = 7$ collision at $j_2 - j_1 = n - 3 = 7$ resolved in the 3×3 top-right corner at $(0, 7)$.

Example 4.4 (Family 2 d-Diagonal Parity Jump: $n = 10$, $d = 0$). Along the main d-diagonal $i - j = 0$ for $n = 10$, the single collision on the entire board occurs between $(0, 0)$ and $(9, 9)$ ($j_2 - j_1 = 9 = n - 1$), which both have $S_{0,0} = S_{9,9} = 6$. The values on $d = 0$ are $\{0, 2, 3, 4, 5, \mathbf{6}, 7, 8, 9\}$, leaving $\mathbf{1}$ vacant. Setting $f(9, 9) = 1 < 6$ produces the full permutation $\{0, \ldots, 9\}$ (Figure 4).

	0	1	2	3	4	5	6	7	8	9
0	**6**	8	0	2	4	9	1	3	5	7
1	7	**9**	1	3	5	0	2	4	6	8
2	8	0	**2**	4	6	1	3	5	7	9
3	9	1	3	**5**	7	2	4	6	8	0
4	0	2	4	6	**8**	3	5	7	9	1
5	1	3	5	7	9	**4**	6	8	0	2
6	2	4	6	8	0	5	**7**	9	1	3
7	3	5	7	9	1	6	8	**0**	2	4
8	4	6	8	0	2	7	9	1	**3**	5
9	5	7	9	1	3	8	0	2	4	**6** ↓1

Cell (i, j)	$S_{i,j}$	$f(i, j)$	Status
$(0, 0)$	6	6	Keeps value
$(1, 1)$	9	9	Distinct
$(2, 2)$	2	2	Distinct
$(3, 3)$	5	5	Distinct
$(4, 4)$	8	8	Distinct
$(5, 5)$	4	4	Distinct
$(6, 6)$	7	7	Distinct
$(7, 7)$	0	0	Distinct
$(8, 8)$	3	3	Distinct
$(9, 9)$	6	**1**	**Steals vacant 1 (< 6)**

Figure 4: Family 2 ($n = 10$): d-diagonal $d = 0$ collision at $j_2 - j_1 = n - 1 = 9$ resolved at $(9, 9)$.

Example 4.5 (Family 3 s-Diagonal Block Jumps: $n = 27$, $s = 25$). For $n = 27$ ($k = 9$), along $i + j = 25$ the step inside each block is $+2$. At the block jump $j = 8 \to 9$ ($u = 0 \to 1$), $S_{i,j}$ jumps from $S_{17,8} = 5$ to $S_{16,9} = 11$, skipping **7** and **9** ($\equiv s + k, s + k + 2 \pmod{27}$). At $j = 17 \to 18$ ($u = 1 \to 2$), $S_{i,j}$ jumps from $S_{8,17} = 0$ to $S_{7,18} = 6$, skipping **2** and **4** ($\equiv s + 4, s + 6 \pmod{27}$). In the 4×4 top-right corner ($j_2 - j_1 = n - 4 = 23$), cells $(2, 23)$, $(1, 24)$, $(0, 25)$ collide with $(25, 0)$, $(24, 1)$, $(23, 2)$ at values $16, 18, 20$. Subtracting $k = 9$ at $j \in \{23, 24\}$ gives $f(2, 23) = 7$ and $f(1, 24) = 9$, and subtracting $2k = 18$ at $j = 25$ gives $f(0, 25) = 2$, filling the exact skipped slots (Figure 5).

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26
0	18	21	24	0	3	6	9	12	15	22	25	1	4	7	10	13	16	19	26	2	5	8	11	14	17	**20** ↓2	23
1	19	22	25	1	4	7	10	13	16	23	26	2	5	8	11	14	17	20	0	3	6	9	12	15	**18** ↓9	21	24
2	20	23	26	2	5	8	11	14	17	24	0	3	6	9	12	15	18	21	1	4	7	10	13	**16** ↓7	19	22	25
3	21	24	0	3	6	9	12	15	18	25	1	4	7	10	13	16	19	22	2	5	8	11	**14**	17	20	23	26
4	22	25	1	4	7	10	13	16	19	26	2	5	8	11	14	17	20	23	3	6	9	**12**	15	18	21	24	0
5	23	26	2	5	8	11	14	17	20	0	3	6	9	12	15	18	21	24	4	7	**10**	13	16	19	22	25	1
6	24	0	3	6	9	12	15	18	21	1	4	7	10	13	16	19	22	25	5	**8**	11	14	17	20	23	26	2
7	25	1	4	7	10	13	16	19	22	2	5	8	11	14	17	20	23	26	**6**	9	12	15	18	21	24	0	3
8	26	2	5	8	11	14	17	20	23	3	6	9	12	15	18	21	24	**0**	7	10	13	16	19	22	25	1	4
9	0	3	6	9	12	15	18	21	24	4	7	10	13	16	19	22	**25**	1	8	11	14	17	20	23	26	2	5
10	1	4	7	10	13	16	19	22	25	5	8	11	14	17	20	**23**	26	2	9	12	15	18	21	24	0	3	6
11	2	5	8	11	14	17	20	23	26	6	9	12	15	18	**21**	24	0	3	10	13	16	19	22	25	1	4	7
12	3	6	9	12	15	18	21	24	0	7	10	13	16	**19**	22	25	1	4	11	14	17	20	23	26	2	5	8
13	4	7	10	13	16	19	22	25	1	8	11	14	**17**	20	23	26	2	5	12	15	18	21	24	0	3	6	9
14	5	8	11	14	17	20	23	26	2	9	12	**15**	18	21	24	0	3	6	13	16	19	22	25	1	4	7	10
15	6	9	12	15	18	21	24	0	3	10	**13**	16	19	22	25	1	4	7	14	17	20	23	26	2	5	8	11
16	7	10	13	16	19	22	25	1	4	**11**	14	17	20	23	26	2	5	8	15	18	21	24	0	3	6	9	12
17	8	11	14	17	20	23	26	2	**5**	12	15	18	21	24	0	3	6	9	16	19	22	25	1	4	7	10	13
18	9	12	15	18	21	24	0	**3**	6	13	16	19	22	25	1	4	7	10	17	20	23	26	2	5	8	11	14
19	10	13	16	19	22	25	**1**	4	7	14	17	20	23	26	2	5	8	11	18	21	24	0	3	6	9	12	15
20	11	14	17	20	23	**26**	2	5	8	15	18	21	24	0	3	6	9	12	19	22	25	1	4	7	10	13	16
21	12	15	18	21	**24**	0	3	6	9	16	19	22	25	1	4	7	10	13	20	23	26	2	5	8	11	14	17
22	13	16	19	**22**	25	1	4	7	10	17	20	23	26	2	5	8	11	14	21	24	0	3	6	9	12	15	18
23	14	17	**20**	23	26	2	5	8	11	18	21	24	0	3	6	9	12	15	22	25	1	4	7	10	13	16	19
24	15	**18**	21	24	0	3	6	9	12	19	22	25	1	4	7	10	13	16	23	26	2	5	8	11	14	17	20
25	**16**	19	22	25	1	4	7	10	13	20	23	26	2	5	8	11	14	17	24	0	3	6	9	12	15	18	21
26	17	20	23	26	2	5	8	11	14	21	24	0	3	6	9	12	15	18	25	1	4	7	10	13	16	19	22

Cell (i, j)	$S_{i,j}$	$f(i, j)$	Status
$(25, 0)$	16	16	Keeps value
$(24, 1)$	18	18	Keeps value
$(23, 2)$	20	20	Keeps value
...	...	...	Block $u = 0$
$(17, 8) \to (16, 9)$	$5 \to 11$	$5 \to 11$	Skips **7**, **9**
$(8, 17) \to (7, 18)$	$0 \to 6$	$0 \to 6$	Skips **2**, **4**
$(2, 23)$	16	**7**	**Steals 7** $(-k)$
$(1, 24)$	18	**9**	**Steals 9** $(-k)$
$(0, 25)$	20	**2**	**Steals 2** $(-2k)$

Figure 5: Family 3 ($n = 27$, $k = 9$): s-diagonal $s = 25$ collisions in the 4×4 top-right corner shifted by $-k$ and $-2k$ into slots skipped at $j = 9, 18$.

Example 4.6 (Family 3 d-Diagonal Block Jumps: $n = 27$, $d = 0$). Along the main d-diagonal $i - j = 0$ for $n = 27$ ($k = 9$), the step inside each block is $+4$. At $j = 8 \to 9$, $S_{i,i}$ jumps from 23 to 4, skipping $\mathbf{0} \equiv d \pmod{27}$; at $j = 17 \to 18$, $S_{i,i}$ jumps from 9 to 17, skipping $\mathbf{13} \equiv d + k + 4 \pmod{27}$. In the 2×2 bottom-right corner ($j_2 - j_1 = n - 2 = 25$), $(25, 25)$ collides with $(0, 0)$ at 18 and steals $f(25, 25) = 18 - 2k = 0$, while $(26, 26)$ collides with $(1, 1)$ at 22 and steals $f(26, 26) = 22 - k = 13$ (Figure 6).

	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26
0	**18**	21	24	0	3	6	9	12	15	22	25	1	4	7	10	13	16	19	26	2	5	8	11	14	17	20	23
1	19	**22**	25	1	4	7	10	13	16	23	26	2	5	8	11	14	17	20	0	3	6	9	12	15	18	21	24
2	20	23	**26**	2	5	8	11	14	17	24	0	3	6	9	12	15	18	21	1	4	7	10	13	16	19	22	25
3	21	24	0	**3**	6	9	12	15	18	25	1	4	7	10	13	16	19	22	2	5	8	11	14	17	20	23	26
4	22	25	1	4	**7**	10	13	16	19	26	2	5	8	11	14	17	20	23	3	6	9	12	15	18	21	24	0
5	23	26	2	5	8	**11**	14	17	20	0	3	6	9	12	15	18	21	24	4	7	10	13	16	19	22	25	1
6	24	0	3	6	9	12	**15**	18	21	1	4	7	10	13	16	19	22	25	5	8	11	14	17	20	23	26	2
7	25	1	4	7	10	13	16	**19**	22	2	5	8	11	14	17	20	23	26	6	9	12	15	18	21	24	0	3
8	26	2	5	8	11	14	17	20	**23**	3	6	9	12	15	18	21	24	0	7	10	13	16	19	22	25	1	4
9	0	3	6	9	12	15	18	21	24	**4**	7	10	13	16	19	22	25	1	8	11	14	17	20	23	26	2	5
10	1	4	7	10	13	16	19	22	25	5	**8**	11	14	17	20	23	26	2	9	12	15	18	21	24	0	3	6
11	2	5	8	11	14	17	20	23	26	6	9	**12**	15	18	21	24	0	3	10	13	16	19	22	25	1	4	7
12	3	6	9	12	15	18	21	24	0	7	10	13	**16**	19	22	25	1	4	11	14	17	20	23	26	2	5	8
13	4	7	10	13	16	19	22	25	1	8	11	14	17	**20**	23	26	2	5	12	15	18	21	24	0	3	6	9
14	5	8	11	14	17	20	23	26	2	9	12	15	18	21	**24**	0	3	6	13	16	19	22	25	1	4	7	10
15	6	9	12	15	18	21	24	0	3	10	13	16	19	22	25	**1**	4	7	14	17	20	23	26	2	5	8	11
16	7	10	13	16	19	22	25	1	4	11	14	17	20	23	26	2	**5**	8	15	18	21	24	0	3	6	9	12
17	8	11	14	17	20	23	26	2	5	12	15	18	21	24	0	3	6	**9**	16	19	22	25	1	4	7	10	13
18	9	12	15	18	21	24	0	3	6	13	16	19	22	25	1	4	7	10	**17**	20	23	26	2	5	8	11	14
19	10	13	16	19	22	25	1	4	7	14	17	20	23	26	2	5	8	11	18	**21**	24	0	3	6	9	12	15
20	11	14	17	20	23	26	2	5	8	15	18	21	24	0	3	6	9	12	19	22	**25**	1	4	7	10	13	16
21	12	15	18	21	24	0	3	6	9	16	19	22	25	1	4	7	10	13	20	23	26	**2**	5	8	11	14	17
22	13	16	19	22	25	1	4	7	10	17	20	23	26	2	5	8	11	14	21	24	0	3	**6**	9	12	15	18
23	14	17	20	23	26	2	5	8	11	18	21	24	0	3	6	9	12	15	22	25	1	4	7	**10**	13	16	19
24	15	18	21	24	0	3	6	9	12	19	22	25	1	4	7	10	13	16	23	26	2	5	8	11	**14**	17	20
25	16	19	22	25	1	4	7	10	13	20	23	26	2	5	8	11	14	17	24	0	3	6	9	12	15	**18** ↓0	21
26	17	20	23	26	2	5	8	11	14	21	24	0	3	6	9	12	15	18	25	1	4	7	10	13	16	19	**22** ↓13

Cell (i, j)	$S_{i,j}$	$f(i, j)$	Status
$(0, 0)$	18	18	Keeps value
$(1, 1)$	22	22	Keeps value
$\dots$	$\dots$	$\dots$	Block $u = 0$
$(8, 8) \to (9, 9)$	$23 \to 4$	$23 \to 4$	Skips **0**
$(17, 17) \to (18, 18)$	$9 \to 17$	$9 \to 17$	Skips **13**
$\dots$	$\dots$	$\dots$	Block $u = 2$
$(24, 24)$	14	14	Distinct
$(25, 25)$	18	**0**	**Steals 0** $(-2k)$
$(26, 26)$	22	**13**	**Steals 13** $(-k)$

Figure 6: Family 3 ($n = 27$, $k = 9$): d-diagonal $d = 0$ collisions in the 2×2 bottom-right corner shifted by $-2k$ and $-k$ into slots skipped at $j = 9, 18$.

Example 4.7 (Family 4 s-Diagonal Seam Shifts: $n = 12$, $s = 9$). For $n = 12$ ($m = 6, k = 4$), $p = (10, 4, 6, 8, 0, 1, 3, 5, 7, 9, 11, 2)$. Along $i + j = 9$, the first main block ($1 \le j \le 3$) visits $\{0, 1, 2\}$ and the second main block ($5 \le j \le 10$) visits $\{5, 6, 7, 8, 9\}$, leaving $\mathbf{3} \equiv s + m \pmod{12}$ and $\mathbf{4} \equiv s + m + 1 \pmod{12}$ vacant between the two blocks. Two seam collisions occur with the pins: $(4, 5)$ ($j = m - 1$) collides with pin $(5, 4)$ ($j = m - 2$) at $S = 5$ and shifts to $f(4, 5) = 5 - 1 = 4$; $(2, 7)$ ($j = n - 5$) collides with pin $(9, 0)$ ($j = 0$) at $S = 7$ and shifts by $-(m - 2) = -4$ to $f(2, 7) = 7 - 4 = 3$ (Figure 7).

	0	1	2	3	4	5	6	7	8	9	10	11
0	10	4	6	8	0	1	3	5	7	**9**	11	2
1	11	5	7	9	1	2	4	6	**8**	10	0	3
2	0	6	8	10	2	3	5	**7** ↓3	9	11	1	4
3	1	7	9	11	3	4	**6**	8	10	0	2	5
4	2	8	10	0	4	**5** ↓4	7	9	11	1	3	6
5	3	9	11	1	**5**	6	8	10	0	2	4	7
6	4	10	0	**2**	6	7	9	11	1	3	5	8
7	5	11	**1**	3	7	8	10	0	2	4	6	9
8	6	**0**	2	4	8	9	11	1	3	5	7	10
9	**7**	1	3	5	9	10	0	2	4	6	8	11
10	8	2	4	6	10	11	1	3	5	7	9	0
11	9	3	5	7	11	0	2	4	6	8	10	1

Cell (i, j)	$S_{i,j}$	$f(i, j)$	Status
$(9, 0)$	7	7	Pin $j = 0$
$(8, 1)$	0	0	Distinct
$(7, 2)$	1	1	Distinct
$(6, 3)$	2	2	Distinct
$(5, 4)$	5	5	Pin $j = m - 2$
$(4, 5)$	5	**4**	**Steals 4 (-1)**
$(3, 6)$	6	6	Distinct
$(2, 7)$	7	**3**	**Steals 3 ($-(m - 2)$)**
$(1, 8)$	8	8	Distinct
$(0, 9)$	9	9	Distinct

Figure 7: Family 4 ($n = 12$, $m = 6, k = 4$): s-diagonal $s = 9$ seam collisions at $j = m - 1 = 5$ and $j = n - 5 = 7$ shifted into vacant slots $\{3, 4\}$.

Example 4.8 (Family 4 d-Diagonal Modular Wrap-Around: $n = 12$, $d = -4$). Along $i-j = -4$ ($i = j-4$) for $n = 12$ ($m = 6, k = 4$), the second main block ($5 \le j \le 10$) steps by $+3$: $S_{1,5} = 2$, $S_{2,6} = 5$, $S_{3,7} = 8$, $S_{4,8} = 11$, all in residue class 2 (mod 3), while pins $(0, 4)$ and $(7, 11)$ have values 0 and 9 (0 (mod 3)), leaving residue class 1 (mod 3) ($\{1, 4, 7, 10\}$) completely vacant. At $j = 9 = m-1+k$ and $j = 10 = m+k$, the progression wraps around 12 to $S_{5,9} = 2 = S_{1,5}$ and $S_{6,10} = 5 = S_{2,6}$. Shifting by -1 at $j = 9$ gives $f(5, 9) = 1$, and shifting by -2 at $j = 10$ gives $f(6, 10) = 3$, resolving both collisions (Figure 8).

	0	1	2	3	4	5	6	7	8	9	10	11
0	10	4	6	8	**0**	1	3	5	7	9	11	2
1	11	5	7	9	1	**2**	4	6	8	10	0	3
2	0	6	8	10	2	3	**5**	7	9	11	1	4
3	1	7	9	11	3	4	6	**8**	10	0	2	5
4	2	8	10	0	4	5	7	9	**11**	1	3	6
5	3	9	11	1	5	6	8	10	0	**2** ↓1	4	7
6	4	10	0	2	6	7	9	11	1	3	**5** ↓3	8
7	5	11	1	3	7	8	10	0	2	4	6	**9**
8	6	0	2	4	8	9	11	1	3	5	7	10
9	7	1	3	5	9	10	0	2	4	6	8	11
10	8	2	4	6	10	11	1	3	5	7	9	0
11	9	3	5	7	11	0	2	4	6	8	10	1

Cell (i, j)	$S_{i,j}$	$f(i, j)$	Status
$(0, 4)$	0	0	Pin $j = m - 2$
$(1, 5)$	2	2	Keeps value
$(2, 6)$	5	5	Keeps value
$(3, 7)$	8	8	Distinct
$(4, 8)$	11	11	Distinct
$(5, 9)$	2	**1**	**Steals 1** (-1)
$(6, 10)$	5	**3**	**Steals 3** (-2)
$(7, 11)$	9	9	Pin $j = n - 1$

Figure 8: Family 4 ($n = 12$, $m = 6, k = 4$): d-diagonal $d = -4$ wrap-around collisions at $j_2 - j_1 = k = 4$ shifted into vacant residue class 1 (mod 3).

References

[1] Mao-Hua Le, Wen Li, and Edward T. Wang, *A generalization of the n-queen problem*, Journal of Systems Science and Mathematical Sciences **3** (1990), no. 2, 183–192, https://sysmath.cjoe.ac.cn/jssc/CN/Y1990/V3/I2/183.